INCH-POUND

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23 April 1991
SUPERSEDING
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MILITARY SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED, DELAY, M31 ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M31 Delay Cartridge Actuated Initiator Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1. Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition, General Specification For
MIL-I-14857	Initiator, Cartridge Actuated, Delay, M10, M31, M42, M72, and M114 Remanufactured Metal Parts Assembly

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, Maryland 20640-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1377

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MIL-I-60296	Initiator, Cartridge Actuated, Delay, M10, M31, M42, M72, and M114 Metal Parts Assembly

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-453	Inspection, Radiographic
MIL-STD-1168	Ammunition Lot Numbering
MS28741	Hose Assembly, Detachable End Fitting, Medium Pressure

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

NAVAL ORDNANCE STATION, INDIAN HEAD, MD (CAGE Codes 19200 and 19203)

8593312	Cartridge, Delay, M93 Assembly
DL8595251	Initiator, Cartridge Actuated, Delay, M31, Assembly
8798142	Box, Packing, Ammunition for Initiator, Cartridge Actuated, Delay, M10, M30A1, M31, M32A1, M42, M43A1, M72, M92, or M114
8798143	Box, Fiberboard (Inner) Packing, Ammunition for Initiator, Cartridge Actuated, Delay, M10, M30A1, M31, M32A1, M42, M43A1, M72, M92, or M114
9216933	Box, Fiberboard (Outer) Alternative, Packing Ammunition, for Initiator, Cartridge Actuated, Delay, M10, M30A1, M31, M32A1, M42, M43A1, M72, M92, or M114

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U.S. AIR FORCE

TPO 00-062-5879 Initiator, Cartridge Actuated, M72

(Application for copies should be addressed to: Commanding Officer, Naval Ordnance Station, Data Control Branch (Code 3720), Indian Head, MD 20640-5000.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the initiators shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2 Conformance to documents. The M31 initiator covered by this specification shall be manufactured in accordance with DL8595251.

3.3 Components.

3.3.1 Primary components. For the purpose of this specification, the M93 delay cartridge (Drawing 8593312) is defined as a primary component. Only primary components from a single lot shall be used in a production lot of initiators. One primary component production lot may be used in more than one initiator production lot.

3.3.2 Component parts. All component parts shall meet the applicable requirements of MIL-I-14857 or MIL-I-60296 prior to assembly.

3.4 Breakaway torque. The chamber and cap of each assembled initiator shall withstand a breakaway torque application of 400 inch-pounds without loosening or deformation (see 4.3.1).

3.5 Residual magnetism. The initiator shall not deflect the indicator of the compass more than five degrees in either direction (see 4.3.2).

3.6 Radiographic examination. Each initiator assembly shall be radiographically examined for defects (see 4.3.3).

3.7 Functional.

3.7.1 Ballistics. Initiators loaded with cartridges from one accepted lot and temperature conditioned shall perform within the pressure values and delay times (see 6.6.1) specified in table I.

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Pressure values shall be recorded at the end of 15 feet of aircraft hose, assembly number MS28741-4, and a pressure chamber with a volume of 1 cubic inch.

TABLE I. Pressure values.

Temperature (°F)	Pressure (psi)		Delay time (sec)
	Minimum	Maximum	
-60 to -70	500	1250	1.00 + 0.30 - 0.20
65 to 75	600	1500	1.00 + 0.30 - 0.20
195 to 205	800	1800	1.00 + 0.30 - 0.20

3.7.1.1 Initiation. The initiator shall be fired by a source delivering an output pressure of 700 +25, -0 pounds per square inch (gage) (psig) at 70 ± 5°F.

3.7.2 Misfires. There shall be no misfires.

3.7.3 Mechanical failures. The initiator shall function without mechanical failure (see 6.6.2).

3.8 Workmanship. Initiators shall be free of cracks, splits, incomplete or damaged threads on inlet and outlet ends, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.8.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.8.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.8.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection

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system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspection. Inspection shall be in accordance with MIL-A-2550 and MIL-STD-453.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with MIL-STD-105.

4.2.1.2 Size of lot. A lot shall consist of an identifiable group of assembled initiators containing metal parts assemblies produced by a homogeneous process and loaded with one lot of accepted cartridges.

4.2.2 Examination. One-hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed in accordance with the classification of defects. Four power (4X) magnification may be used for inspection of such defects as burr and foreign matter. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.2.2.1 Cartridge, Delay, M93, Drawing 8593312, covering a detail of Drawing 8595251.

Characteristic	Defect	Method of Inspection
Critical		
1	Mixed cartridges	Visual
2	Metal defective	Visual
3	O-ring missing	Visual
Major		
101	O-ring split, cut, or cracked	Visual
102	Foreign matter present	Visual
Minor		
201	Case scratched, wrinkled, bulged, or dented	Visual
202	Varnish missing or inadequate	Visual

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4.2.2.2 Assembly, Drawing 8595251, less Drawings 8595254, 8594857, 8594860, 8593312, and MIL-C-5501/10, and MIL-C-5501/11.

Characteristic	Defect	Method of Inspection
Critical		
1	Set screw missing	Visual
Major		
101	Thread damaged	Visual
102	Burr	Visual
103	Foreign matter present	Visual
Minor		
None defined		

4.2.2.3 Assembly, Drawing 8595251, less Drawing 8594857, MIL-C-5501/10, and MIL-C-5501/11.

Critical		
1	Cartridge loose	Manual/Visual
2	Cartridge retainer loose	Manual/Visual
3	Metering groove of cartridge retainer missing	Visual
4	Elongated hole of cartridge retainer missing	Visual
Major		
None defined		
Minor		
None defined		

4.2.2.4 Assembly, Drawings 8595251, 10521631, less MIL-C-5501/10, and MIL-C-5501/11.

Critical		
1	Location of cartridge retainer	X-ray
2	Shear pin not engaged in firing pin	Gage/X-ray
3	Any part missing	Visual/X-ray
4	Improper assembly	Visual/X-ray
5	Foreign matter in gas port	Visual
Major		
101	Length from firing pin housing boss face to gas port face, max	Gage

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Characteristic	Defect	Method of Inspection
Minor		
201	Varnish missing or inadequate	Visual
202	Location of label, incorrect	Visual
203	Barrier material missing from set screw	Visual

4.2.2.5 Assembly, Drawing 8595251.

Critical		
1	Marking missing, incorrect or illegible	Visual
Major		
101	Height of cap, max	Gage
102	Cap set screw missing, loose or above flush	Visual
Minor		
201	Total length, max	Measure
202	Location of marking incorrect	Visual
203	Protective cap or plug missing	Visual
204	Protective finish inadequate or defective	Visual

4.2.2.6 Box, Fiberboard (Inner) Ammunition, Drawing 8798143.

Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual
202	Box damaged	Visual
203	Sealing strip broken or missing	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Cut, damaged, or improperly sealed bag	Visual

4.2.2.7 Box, Ammunition Packing, Drawing 8798142.

Critical		
None defined		
Major		
None defined		

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Characteristic	Defect	Method of Inspection
Minor		
201	Contents loose	Manual
202	Strapping missing, loose, broken, or improperly engaged	Visual/Manual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.2.2.8 Box, Fiberboard (Outer) Alternative Ammunition Packing, Drawing 9216933.

Critical
None defined

Major
None defined

Minor		
201	Contents loose	Manual/Visual
202	Carton broken or damaged	Visual
203	Number of units contained incorrect	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

4.2.3 Inspection testing. The following tests shall be performed in accordance with 4.3.

4.2.3.1 Breakaway torque. Each assembly shall be subjected to the breakaway torque test of the threaded joint between the cap and the chamber. Failure of any assembly to comply with 3.4 shall be cause for rejection of the assembly.

4.2.3.2 Residual magnetism. Each initiator shall be subjected to the test for excessive residual magnetism as specified in 4.3.2. Failure of any initiator to comply with 3.5 shall be cause for rejection of the initiator.

4.2.3.3 Functional tests. The following tests shall be performed by the Government.

4.2.3.3.1 Ballistics. Twenty-one units from each lot shall be tested to determine compliance with 3.7. The units shall be temperature conditioned as provided in table II and 4.3.4.2. Failure of any unit to comply with 3.7 shall be cause for rejection of the lot.

4.2.3.3.2 Retest. There shall be no retests.

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TABLE II. Conditioning temperature.

Temperature (°F)	Sample size
-60 to -70	7
65 to 75	7
195 to 205	7

4.2.3.3.3 Test Failure. If a test failure is attributable to an assignable cause, excluding the test initiator, the original test results shall be discarded and that test reconducted.

4.3 Test methods and procedures.

4.3.1 Breakaway torque. A torque wrench set as specified in 3.4 shall be used in conjunction with a fixture to determine the adequacy of the joints. The chamber shall be secured in an air vise, and the torque shall be applied to the cap.

4.3.2 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

4.3.3 Radiographic examination. All assembled initiators shall be radiographically examined in accordance with MIL-STD-453, with exception that an initiator assembly will be used for the penetrameter. Any observable imperfections in components or assembly shall be cause for rejection. The initiators shall be positioned for the most revealing exposure, with the long axis of each assembly perpendicular to the rays of the X-ray machine. All initiators shall be identified with serial numbers prior to X-ray examination. The serial numbers shall be in consecutive numerical order beginning with the number 001 in each production lot. Any missing serial numbers are to be identified on the X-rays. The initiators shall be arranged on boards or trays in consecutive order and each radiograph shall carry a permanent identification of the units displayed thereon. The radiographic plate identification shall include item nomenclature, the complete lot number, the span of serial numbers displayed, and the contract number. The radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test. Defective initiators found during X-ray review are to be marked on the X-ray prior to shipment and those defective items shall be removed from the production lot. Film interpretation of the radiographs shall verify the presence and correct positioning of the following parts:

a. Cartridge, Delay M93 P/N 8593312
b. Firing pin P/N 8594744
c. Chamber P/N 8594857

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d.	Firing pin housing	P/N 8594858
e.	Cartridge retainer	P/N 8594860
f.	CAP	P/N 8595254
g.	O-ring	MS28775-009
h.	Setscrew, Shear Pin	MS51029-1
i.	Setscrew, Cap (2)	Commercial
j.	O-ring	MS28775-214

4.3.4 Functional.

4.3.4.1 Ballistics. The test sample shall be rigidly supported in the fixture and fired by a source delivering an output of 700 + 25, -0 psig at the end of 3 feet of aircraft hose, MS28741-4, at 70 ± 5°F. The sample shall be tested within 5 minutes after removal from the temperature conditioning chamber. When this time limit is exceeded, the sample units shall be reconditioned in accordance with 4.3.4.2. Required pressure readings shall be taken as specified in 3.7.1.

4.3.4.2 Temperature conditioning. Each assembled initiator shall be conditioned for a period of time not less than 4 hours nor greater than 24 hours. When reconditioning initiators at -60°F, all condensation shall be removed from the interior surfaces before returning the initiator to the conditioning chamber.

4.4 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with section 5.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be levels A or B as specified by the contracting activity:

Level A - In accordance with Drawings 8798142 and 8798143
Level B - In accordance with Drawings 8798143 and 9216933

5.1.1 Lot numbers. Lot numbers shall be in accordance with MIL-STD-1168.

5.2 Air Force packaging. Air Force packaging shall be in accordance with TPO 00-062-5879.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The initiator assembly is used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

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6.2. Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Production test activity and lot size (see 4.1 and 4.2.1).

d. Direction for shipping radiographs of entire production lot along with ballistic sample to the activity conducting production lot acceptance tests (see 4.3.3).

e. Level of marking and packing (see 5.1).

f. Safety precautions (see 6.4).

6.3 Explosive hazard classification information.

Hazard Classification/Division: 1.4
Storage Compatibility Group: S
DOT Class: C
DOT Marking - Explosive Power Device - Class C - Handle Carefully - Keep Fire Away

6.4 Safety precautions. The safety precaution requirements of "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) 23.3. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

6.5 Hazard notice. The initiator described herein and certain of its components are flammable, explosive, or both. Consequently, they present hazards in manufacture, handling, storage, and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically-induced incidents.

6.6 Definitions.

6.6.1 Delay time. Delay time is the time from primer impact to the start of pressure rise.

6.6.2 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

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6.7 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian:
Navy-OS
Air Force-99

Review activity:
Air Force-70

Preparing activity:
Navy-OS
(Project 1377-ND68)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-I-60639A	11 MARCH 1991

3. DOCUMENT TITLE
INITIATOR, CARTRIDGE ACTUATED, DELAY, M31 ASSEMBLY

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
COMMANDING OFFICER NAVAL ORDNANCE STATION (CODE 3730)	301-743-4358/4510	364-4358/4510
c. ADDRESS (Include Zip Code) INDIAN HEAD, MD 20640-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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